INTERNET PROGRAMMING LAB

Experiment No. 9

Title: Online Shopping Website for Electronic Gadgets (TechSavvy)

AIM

To design and develop an online shopping website using HTML, CSS and JavaScript that displays categories and sub-categories of electronic gadgets, supports searching, filtering, shopping cart, wish list, coupon discount, checkout and order tracking, and allows the administrator to add/remove items, change price and description and view customer checkout details.

ALGORITHM

Step 1. Start and display the Home page with all categories, sub-categories, popular recommendations and the product list.

Step 2. Search / Filter: read the search text, selected category, sub-category, maximum price and sort option; filter and sort the product array and redisplay.

Step 3. Details: on clicking Details display description, availability and customer reviews of the item.

Step 4. Wish list: if the user is logged in, add or remove the item from the wish list.

Step 5. Add to Cart: add the item (or increase quantity up to available stock) and update the cart count.

Step 6. Cart: display items with price, quantity and subtotal; allow editing quantity, removing items, and compute the total.

Step 7. Checkout: ensure the user is logged in; read name, address, payment method and coupon; apply 10% discount for coupon TECH10.

Step 8. Create the order with status "Processing", reduce stock, clear the cart and show an order confirmation.

Step 9. Orders page: display order history and current status of the logged-in customer.

Step 10. Admin login: display all items; the admin can add a new item, remove an item, modify price and description.

Step 11. Admin views the customer details of every checkout and can update the order status (Processing / Shipped / Delivered).

Step 12. Logout and stop.

PROGRAM

File: p9.html (HTML structure, CSS styling and JavaScript logic in a single file)

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>TechSavvy - Online Gadget Store</title>
<style>
:root{--c:#5b2a86}
*{box-sizing:border-box;margin:0;padding:0;font-family:Segoe UI,Arial,sans-serif}
body{background:#f2f5fa;color:#222;line-height:1.4}
header{background:var(--c);color:#fff;padding:12px 20px;display:flex;justify-content:space-between;align-
items:center;flex-wrap:wrap;gap:8px}
nav button{background:none;border:0;color:#fff;margin-left:4px;padding:7px 11px;cursor:pointer;border-
radius:4px;font-size:14px}
nav button:hover{background:#ffffff33}
main{max-width:1000px;margin:18px auto;padding:0 12px}
.card{background:#fff;border-radius:8px;padding:16px;margin-bottom:16px;box-shadow:0 1px 4px #0002}
.page{display:none}.page.on{display:block}
h3{margin-bottom:10px;color:var(--c)}
input,select,textarea{padding:7px;margin:4px 6px 4px 0;border:1px solid #bbb;border-radius:4px;font-
size:14px}
.btn{background:var(--c);color:#fff;border:0;padding:8px 14px;border-radius:4px;cursor:pointer;font-
size:14px;margin:2px}
.btn.red{background:#b3261e}.btn.gray{background:#666}
table{width:100%;border-collapse:collapse;margin-top:8px;font-size:14px}
th,td{border:1px solid #ddd;padding:7px;text-align:left}th{background:#e9eef6}
.ok{color:#0a7a2f;margin:6px 0}.err{color:#b3261e;margin:6px 0}
.grid{display:grid;grid-template-columns:repeat(auto-fill,minmax(215px,1fr));gap:12px}
.item{border:1px solid #e1e4ea;border-radius:8px;padding:12px;background:#fff}
.pic{height:90px;border-radius:6px;display:flex;align-items:center;justify-content:center;font-
size:44px;background:linear-gradient(135deg,#dfe9f7,#f7e9df);margin-bottom:8px}
.pill{display:inline-block;background:#e5ecf8;border-radius:12px;padding:2px 10px;font-size:12px;margin:2px}
.stat{display:inline-block;min-width:150px;background:#eef3fb;border-radius:8px;padding:10px
14px;margin:4px;text-align:center}
.stat b{display:block;font-size:22px;color:var(--c)}
@media(max-width:600px){nav button{padding:5px 6px;font-size:12px}table{font-size:12px}}

.price{font-weight:bold;color:#5b2a86;font-size:17px}
.old{text-decoration:line-through;color:#888;font-size:13px}
.cat{cursor:pointer}.cat.on{background:#5b2a86;color:#fff}
.wish{cursor:pointer;font-size:20px;float:right;color:#c33}
</style>
</head>
<body>
<header>
  <h2>&#128241; TechSavvy</h2>
  <div><input id="q" placeholder="Search products..." oninput="renderHome()" style="width:200px"></div>
  <nav>
    <button onclick="go('home')">Home</button>
    <button onclick="go('wish')">Wish List</button>
    <button onclick="go('cart')">Cart (<span id="cc">0</span>)</button>
    <button onclick="go('orders')">Orders</button>
    <button onclick="go('admin')">Admin</button>
    <button id="authBtn" onclick="authClick()">Login</button>
  </nav>
</header>
<main>
  <section id="login" class="page card">
    <h3>Login / Register</h3>
    <input id="lu" placeholder="Username"><input id="lp" type="password" placeholder="Password">
    <button class="btn" onclick="login()">Login</button><button class="btn gray"
onclick="reg()">Register</button>
```

```
    <p id="lmsg"></p><small>Admin demo login: admin / admin123</small>
  </section>

  <section id="home" class="page">
    <div class="card"><h3>Shop by Category</h3><div id="cats"></div><div id="subs" style="margin-
top:6px"></div>
      Max price <input type="number" id="maxp" placeholder="e.g. 50000" style="width:110px"
oninput="renderHome()">
      Sort <select id="sort" onchange="renderHome()"><option value="pop">Popularity</option><option
value="lo">Price: low to high</option><option value="hi">Price: high to low</option></select>
      <span class="pill" style="background:#ffe9b3">Online offer: use coupon TECH10 for 10% off</span>
    </div>
    <div class="card" id="rec"></div>
    <div class="card"><h3 id="ptitle">Products</h3><div id="plist" class="grid"></div></div>
    <div class="card hidden" id="detail" style="display:none"></div>
  </section>

  <section id="wish" class="page card"><h3>My Wish List</h3><div id="wlist" class="grid"></div></section>

  <section id="cart" class="page card">
    <h3>Shopping Cart</h3><div id="clist"></div>
    <div id="co" style="display:none;margin-top:10px">
      <input id="cname" placeholder="Full name"><input id="cadd" placeholder="Shipping address"
style="width:250px">
      <select id="cpay"><option>Credit/Debit Card</option><option>Net
Banking</option><option>UPI</option><option>Cash on Delivery</option></select>
      <input id="cpn" placeholder="Coupon code" style="width:110px">
      <button class="btn" onclick="checkout()">Place Order</button>
    </div>
    <p id="cmsg"></p>
  </section>

  <section id="orders" class="page card"><h3>Order Status &amp;amp; History</h3><div id="olist"></div></section>

  <section id="admin" class="page card">
    <h3>Admin Panel - Manage Items</h3>
    <input id="an" placeholder="Item name"><select id="ac"></select><input id="asub" placeholder="Sub-
category" style="width:120px">
    <input id="ap" type="number" placeholder="Price" style="width:90px"><input id="as" type="number"
placeholder="Stock" style="width:70px">
    <input id="ad" placeholder="Description" style="width:200px">
    <button class="btn" onclick="addItem()">Add Item</button><p id="amsg"></p>
    <div id="atable"></div>
    <h3 style="margin-top:14px">Customer Checkout Information</h3><div id="ctable"></div>
  </section>
</main>

<script>
const $ = id => document.getElementById(id);
const load = (k, d) => { try { return JSON.parse(localStorage.getItem(k)) || d; } catch (e) { return d; } };
const save = (k, v) => { try { localStorage.setItem(k, JSON.stringify(v)); } catch (e) {} };

const CATS = {Smartphones:['Android','iPhone'], Laptops:['Gaming','Ultrabook'], Tablets:['10 inch','Mini'],
Accessories:['Audio','Chargers']};
let users  = load('ts_users', [{u:'admin', p:'admin123', role:'admin'}]);
let items  = load('ts_items', [
  {id:1, name:'Nova X5 Smartphone',  cat:'Smartphones', sub:'Android',  price:24999, stock:12, pop:95,
icon:'&#128241;', desc:'6.5-inch AMOLED, 128GB, 50MP camera.', reviews:['Great display!','Battery lasts 2
days.']},
  {id:2, name:'Apex iPhone 15',      cat:'Smartphones', sub:'iPhone',   price:69999, stock:5,  pop:99,
icon:'&#128241;', desc:'A16 chip, 256GB, dual camera.', reviews:['Super smooth.']},
  {id:3, name:'Titan G7 Gaming Laptop', cat:'Laptops',  sub:'Gaming',   price:89999, stock:4,  pop:80,
icon:'&#128187;', desc:'i7, 16GB RAM, RTX graphics.', reviews:['Runs every game.']},
  {id:4, name:'AirBook 14 Ultrabook',cat:'Laptops',     sub:'Ultrabook',price:54999, stock:9,  pop:88,
icon:'&#128187;', desc:'Lightweight 1.2kg, 12h battery.', reviews:['Perfect for office.','A bit pricey.']},
  {id:5, name:'TabPro 10',           cat:'Tablets',     sub:'10 inch',  price:21999, stock:10, pop:70,
icon:'&#128433;', desc:'10.4-inch display with stylus.', reviews:[]},
  {id:6, name:'BassBuds Wireless Earbuds', cat:'Accessories', sub:'Audio', price:2499, stock:40, pop:92,
icon:'&#127911;', desc:'ANC, 24h playback.', reviews:['Good bass.']},
```

```
  {id:7, name:'FastCharge 65W Adapter', cat:'Accessories', sub:'Chargers', price:1299, stock:25, pop:75,
icon:'&#128268;', desc:'GaN charger with 2 ports.', reviews:[]}
]);
let orders = load('ts_orders', []), wish = load('ts_wish', []), cart = [], me = null, catSel = 'All', subSel
= '';

function msg(id, t, bad) { $(id).className = bad ? 'err' : 'ok'; $(id).textContent = t; }
function go(p) {
  if (['orders', 'wish'].includes(p) && !me) { msg('lmsg', 'Please login first.', 1); p = 'login'; }
  if (p === 'admin' && (!me || me.role !== 'admin')) { msg('lmsg', 'Admin login required.', 1); p = 'login';
}
  document.querySelectorAll('.page').forEach(s => s.classList.remove('on')); $(p).classList.add('on');
  if (p === 'home') renderHome(); if (p === 'cart') renderCart(); if (p === 'orders') renderOrders();
  if (p === 'admin') renderAdmin(); if (p === 'wish') renderWish();
}
function reg() {
  const u = $('lu').value.trim(), p = $('lp').value;
  if (!u || p.length < 4) return msg('lmsg', 'Username and 4+ char password required.', 1);
  if (users.some(x => x.u === u)) return msg('lmsg', 'User exists.', 1);
  users.push({u, p, role:'customer'}); save('ts_users', users); msg('lmsg', 'Registered. Please login.');
}
function login() {
  const f = users.find(x => x.u === $('lu').value.trim() && x.p === $('lp').value);
  if (!f) return msg('lmsg', 'Invalid credentials.', 1);
  me = f; $('authBtn').textContent = 'Logout (' + f.u + ')'; go(f.role === 'admin' ? 'admin' : 'home');
}
function authClick() { if (me) { me = null; cart = []; $('cc').textContent = 0; $('authBtn').textContent =
'Login'; } go('login'); }

/* ---------- catalogue ---------- */
function renderHome() {
  $('cats').innerHTML = ['All', ...Object.keys(CATS)].map(c => `<button class="btn gray cat ${c === catSel ?
'on' : ''}" onclick="catSel='${c}';subSel='';renderHome()">${c}</button>`).join('');
  $('subs').innerHTML = catSel !== 'All' ? CATS[catSel].map(s => `<span class="pill cat ${s === subSel ?
'on' : ''}" onclick="subSel='${s}';renderHome()">${s}</span>`).join('') : '<small>Sub-categories: ' +
Object.values(CATS).flat().join(' &bull; ') + '</small>';
  const q = $('q').value.toLowerCase(), mp = +$('maxp').value || Infinity;
  let list = items.filter(i => (catSel === 'All' || i.cat === catSel) && (!subSel || i.sub === subSel) &&
i.price <= mp && i.name.toLowerCase().includes(q));
  const s = $('sort').value;
  list.sort((a, b) => s === 'lo' ? a.price - b.price : s === 'hi' ? b.price - a.price : b.pop - a.pop);
  $('ptitle').textContent = 'Products (' + list.length + ')';
  $('plist').innerHTML = list.map(card).join('') || '<p>No items match.</p>';
  const last = orders.filter(o => me && o.user === me.u).pop();
  const rc = last ? items.filter(i => i.cat === last.lines[0].cat).slice(0, 3) : [...items].sort((a, b) =>
b.pop - a.pop).slice(0, 3);
  $('rec').innerHTML = `<h3>${last ? 'Recommended based on your purchases' : 'Popular right now'}</h3><div
class="grid">${rc.map(card).join('')}</div>`;
}
function card(i) {
  return `<div class="item"><span class="wish" onclick="toggleWish(${i.id})">${wish.includes(i.id) ?
'&#9829;' : '&#9825;'}</span><div class="pic">${i.icon}</div>
    <b>${i.name}</b><br><span class="pill">${i.cat} / ${i.sub}</span><br><span class="price">Rs. $
{i.price}</span><br>
    <small>${i.stock > 0 ? 'In stock: ' + i.stock : '<span class="err">Out of stock</span>'}</small><br>
    <button class="btn" ${i.stock < 1 ? 'disabled' : ''} onclick="addCart(${i.id})">Add to Cart</button>
    <button class="btn gray" onclick="showDetail(${i.id})">Details</button></div>`;
}
function showDetail(id) {
  const i = items.find(x => x.id === id), d = $('detail');
  d.style.display = 'block';
  d.innerHTML = `<h3>${i.name}</h3><p>${i.desc}</p><p><b>Availability:</b> ${i.stock > 0 ? i.stock + '
units' : 'Out of stock'}</p>
    <b>Customer reviews</b><ul style="margin-left:20px">${i.reviews.map(r => `<li>${r}</li>`).join('') ||
'<li>No reviews yet</li>'}</ul>
    <button class="btn gray" onclick="this.parentNode.style.display='none'">Close</button>`;
  d.scrollIntoView();
}
function toggleWish(id) {
```

```
  if (!me) { msg('lmsg', 'Login to use the wish list.', 1); return go('login'); }
  wish = wish.includes(id) ? wish.filter(x => x !== id) : [...wish, id]; save('ts_wish', wish);
renderHome();
}
function renderWish() { $('wlist').innerHTML = items.filter(i => wish.includes(i.id)).map(card).join('') ||
'<p>Your wish list is empty.</p>'; }

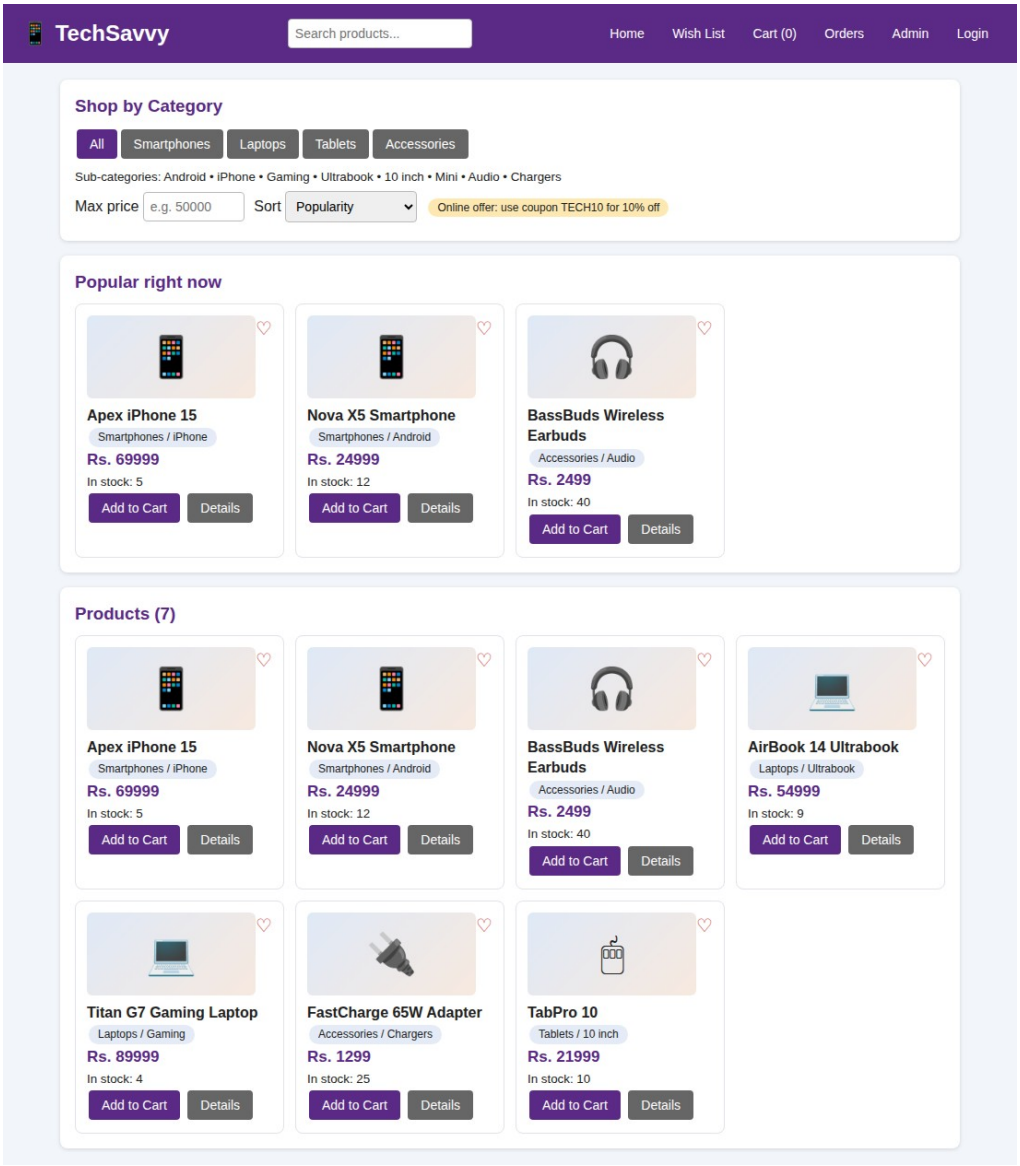
/* ---------- cart ---------- */
function addCart(id) {
  const i = items.find(x => x.id === id), l = cart.find(c => c.id === id);
  if (l) { if (l.qty < i.stock) l.qty++; } else cart.push({id, qty:1});
  $('cc').textContent = cart.reduce((s, c) => s + c.qty, 0);
}
function renderCart() {
  if (!cart.length) { $('clist').innerHTML = '<p>Your cart is empty.</p>'; $('co').style.display = 'none';
return; }
  let tot = 0;
  $('clist').innerHTML = '<table><tr><th>Item</th><th>Price</th><th>Qty</th><th>Subtotal</th><th></th></tr>'
+ cart.map(c => {
    const i = items.find(x => x.id === c.id); tot += i.price * c.qty;
    return `<tr><td>${i.name}</td><td>${i.price}</td><td><input type="number" min="1" max="${i.stock}"
value="${c.qty}" style="width:60px" onchange="setQty(${c.id},this.value)"></td>
    <td>${i.price * c.qty}</td><td><button class="btn red"
onclick="rmCart(${c.id})">Remove</button></td></tr>`;
  }).join('') + `<tr><th colspan="3">Total</th><th colspan="2">Rs. ${tot}</th></tr></table>`;
  $('co').style.display = 'block';
}
function setQty(id, v) { const c = cart.find(x => x.id === id), i = items.find(x => x.id === id); c.qty =
Math.max(1, Math.min(+v || 1, i.stock)); $('cc').textContent = cart.reduce((s, c) => s + c.qty, 0);
renderCart(); }
function rmCart(id) { cart = cart.filter(c => c.id !== id); $('cc').textContent = cart.reduce((s, c) => s +
c.qty, 0); renderCart(); }
function checkout() {
  if (!me) { msg('lmsg', 'Please login to check out.', 1); return go('login'); }
  if (!$('cname').value.trim() || !$('cadd').value.trim()) return msg('cmsg', 'Enter name and shipping
address.', 1);
  let tot = 0; const lines = cart.map(c => { const i = items.find(x => x.id === c.id); i.stock -= c.qty; tot
+= i.price * c.qty; return {name:i.name, cat:i.cat, qty:c.qty, price:i.price}; });
  let disc = 0; if ($('cpn').value.trim().toUpperCase() === 'TECH10') disc = Math.round(tot * 0.1);
  const o = {id:'ORD' + Date.now().toString().slice(-6), user:me.u, name:$('cname').value, addr:$
('cadd').value, pay:$('cpay').value, lines, total:tot - disc, disc, status:'Processing'};
  orders.push(o); save('ts_orders', orders); save('ts_items', items); cart = []; $('cc').textContent = 0;
renderCart();
  msg('cmsg', `Order ${o.id} placed successfully! Amount paid Rs. ${o.total}` + (disc ? ` (discount Rs. $
{disc})` : '') + '. Confirmation sent.');
}
function renderOrders() {
  const mine = orders.filter(o => o.user === me.u);
  $('olist').innerHTML = mine.length ? '<table><tr><th>Order</th><th>Items</th><th>Total</th><th>Payment</
th><th>Status</th></tr>' +
    mine.map(o => `<tr><td>${o.id}</td><td>${o.lines.map(l => l.name + ' x' + l.qty).join(', ')}</td><td>Rs.
${o.total}</td><td>${o.pay}</td><td>${o.status}</td></tr>`).join('') + '</table>' : '<p>No orders yet.</p>';
}

/* ---------- admin ---------- */
function renderAdmin() {
  $('ac').innerHTML = Object.keys(CATS).map(c => `<option>${c}</option>`).join('');
  $('atable').innerHTML = '<table><tr><th>Item</th><th>Category</th><th>Price</th><th>Description</
th><th>Stock</th><th>Actions</th></tr>' + items.map(i =>
    `<tr><td>${i.name}</td><td>${i.cat}/${i.sub}</td><td><input type="number" id="p${i.id}" value="$
{i.price}" style="width:90px"></td>
    <td><input id="d${i.id}" value="${i.desc}" style="width:190px"></td><td>${i.stock}</td>
    <td><button class="btn" onclick="upd(${i.id})">Update</button><button class="btn red" onclick="del($
{i.id})">Remove</button></td></tr>`).join('') + '</table>';
  $('ctable').innerHTML = orders.length ?
'<table><tr><th>Order</th><th>Customer</th><th>Address</th><th>Total</th><th>Status</th></tr>' +
    orders.map(o => `<tr><td>${o.id}</td><td>${o.name} (${o.user})</td><td>${o.addr}</td><td>${o.total}</td>
```

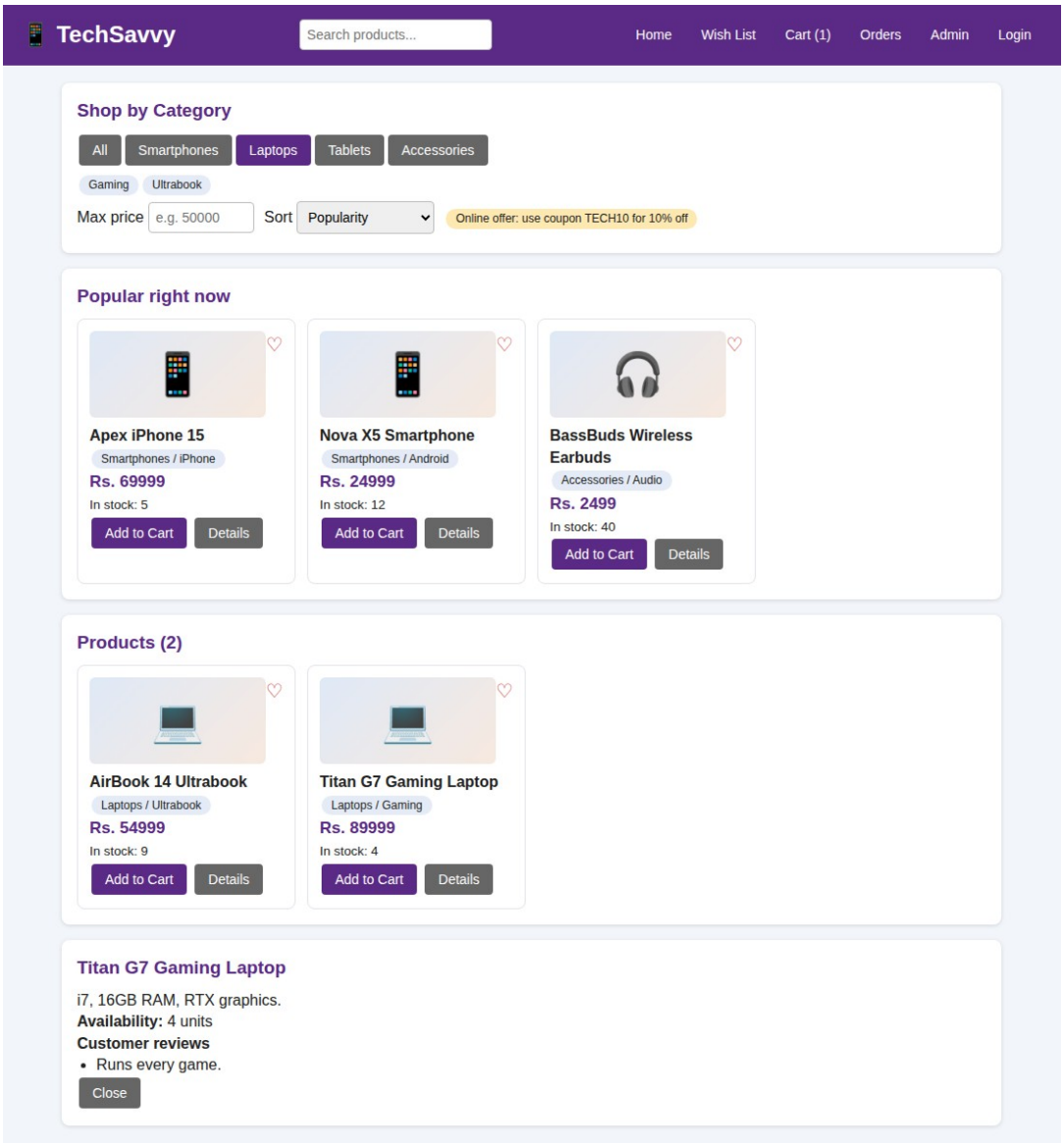
```
    <td><select onchange="setStatus('${o.id}',this.value)">${['Processing','Shipped','Delivered'].map(s =>
`<option ${s === o.status ? 'selected' : ''}>${s}</option>`).join('')}</select></td></tr>`).join('') +
'</table>' : '<p>No customer checkouts yet.</p>';
}
function addItem() {
  const it = {id:Date.now(), name:$('an').value.trim(), cat:$('ac').value, sub:$('asub').value.trim() ||
'General', price:+$('ap').value, stock:+$('as').value, pop:50, icon:'&#128230;', desc:$('ad').value,
reviews:[]};
  if (!it.name || !it.price) return msg('amsg', 'Name and price required.', 1);
  items.push(it); save('ts_items', items); renderAdmin(); msg('amsg', 'Item added.');
}
function upd(id) { const i = items.find(x => x.id === id); i.price = +$('p' + id).value; i.desc = $('d' +
id).value; save('ts_items', items); msg('amsg', 'Price/description updated.'); }
function del(id) { items = items.filter(i => i.id !== id); save('ts_items', items); renderAdmin();
msg('amsg', 'Item removed.'); }
function setStatus(id, s) { orders.find(o => o.id === id).status = s; save('ts_orders', orders); }

go('home');
</script>
</body>
</html>
```

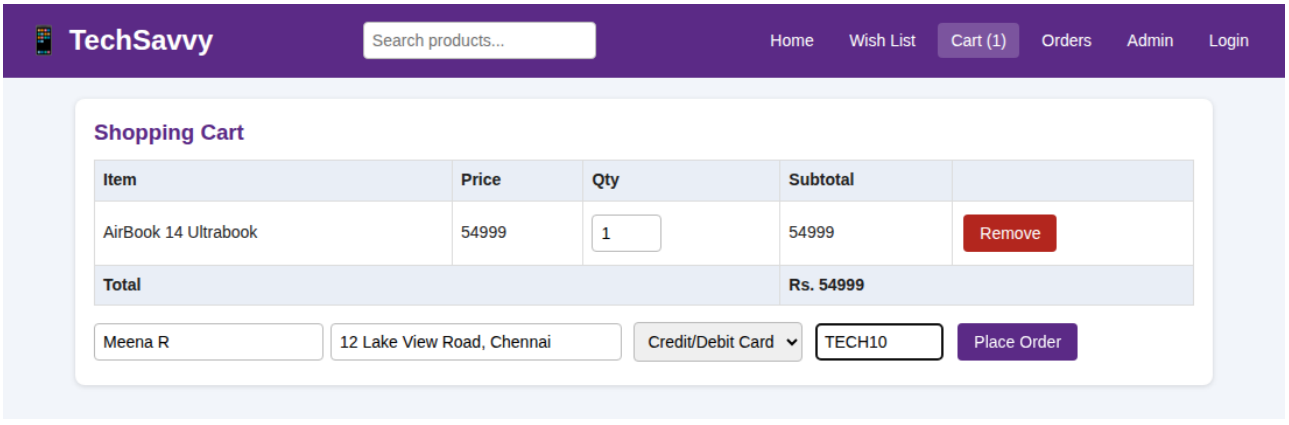
OUTPUT



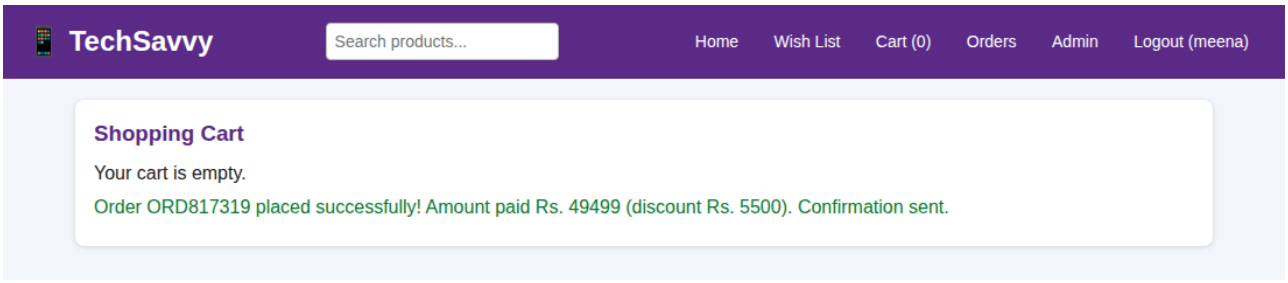
Home page - categories, sub-categories, popular items and product list



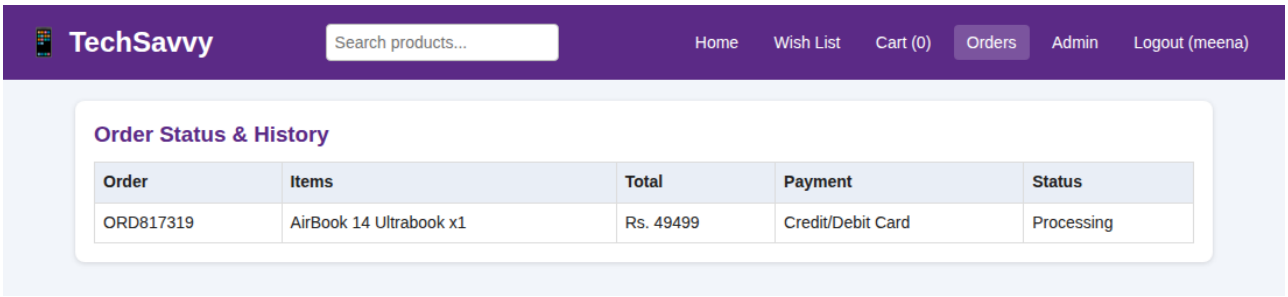
Category filter (Laptops) with item details and reviews



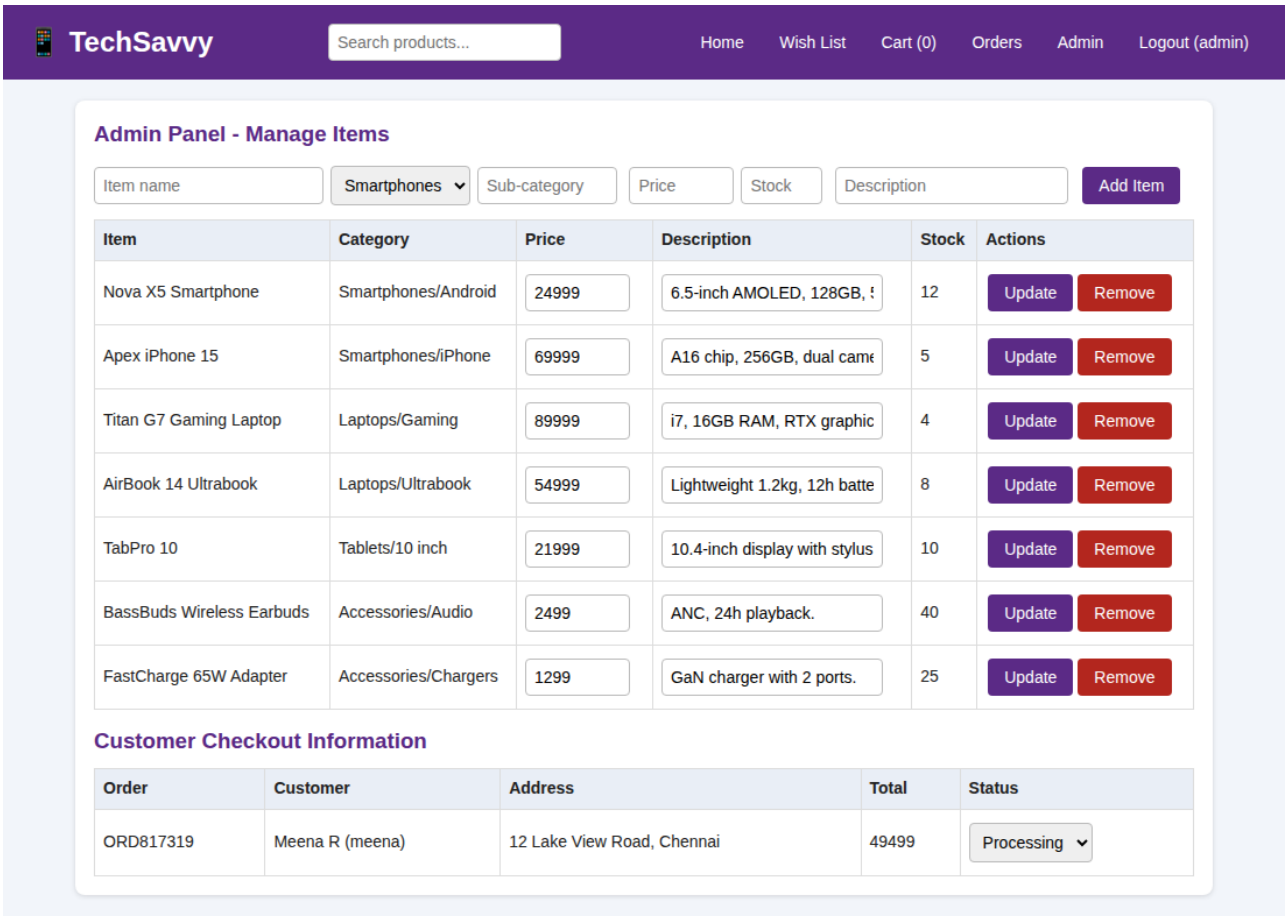
Shopping cart with quantity edit, remove and checkout form



Order placed successfully with coupon discount



Order status and history page



Admin panel - add/remove items, update price/description, customer checkout details

RESULT

Thus the TechSavvy online shopping website was designed and implemented using HTML, CSS and JavaScript. Products were browsed, filtered and added to the cart, an order was placed and tracked, and the administrator managed items and viewed customer information. The output was verified successfully.